CS2201 Computer Organization & Architecture

L3: Review of Floating-Point Representation



John Jose

Associate Professor

Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

Integers and fixed-point numbers

- **Integers: the universe is infinite but discrete**
  - No numbers between consecutive integers, e.g., 5 and 6
  - A countable (finite) number of items in a finite range
  - Referred to as fixed-point numbers

Real Numbers and floating-point numbers

- **Real numbers – the universe is infinite and continuous**
  - Fractions represented by decimal notation
    - Rational numbers, e.g., 5/2 = 2.5
    - Irrational numbers, e.g., 22/7 = 3.14159265 . . .
  - Infinite numbers exist even in the smallest range
  - Referred to as floating-point numbers
- A large number: **976,000,000,000,000 = 9.76×10^{14}**
- A small number: **0.0000000000000976 = 9.76×10^{-14}**

Standard Scientific Notation

- Decimal numbers
  - 0.513×10^5, 5.13×10^4 and 51.3×10^3 are written in scientific notation.
  - **5.13×10^4 is the normalized scientific notation.**
- Binary numbers
  - Base 2
  - Binary point – multiplication by 2 moves the point to the right.
  - **Normalized scientific notation, e.g., $1.0_2 \times 2^{-1}$**

Floating Point Numbers

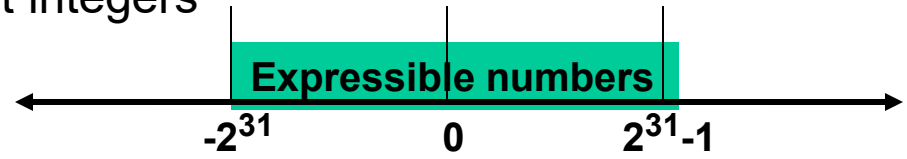
- General format : $\pm 1.bbbbb_2 \times 2^{eeee}$ or $(-1)^S \times (1+F) \times 2^E$

  where

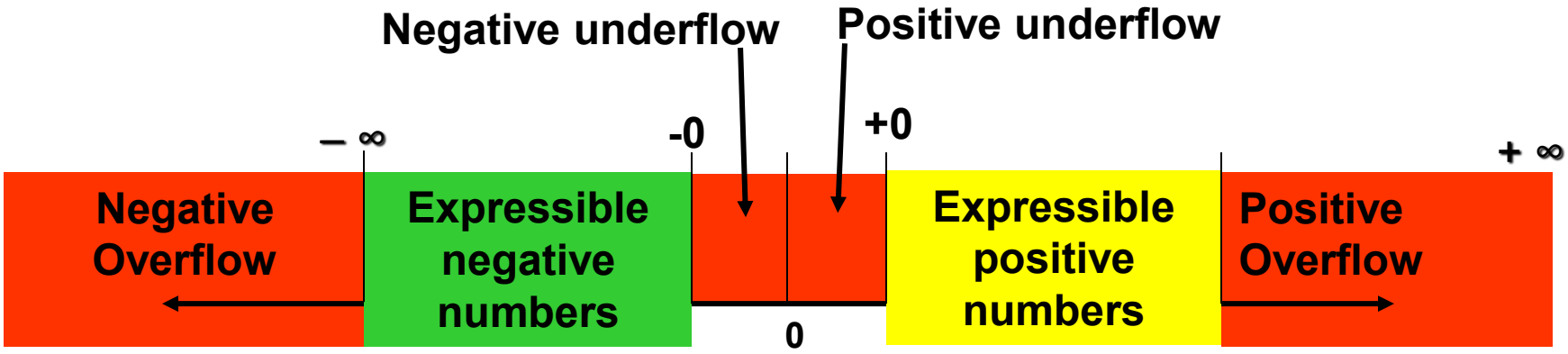
  - S = sign, 0 for positive, 1 for negative
  - F = fraction (or mantissa) as a binary integer, 1+F is called significand
  - E = exponent as a binary integer, positive or negative (two's complement)

Numbers in 32-bit Formats

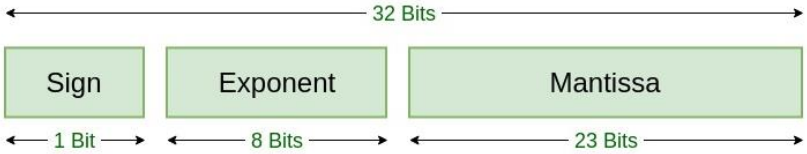
- Two's complement integers



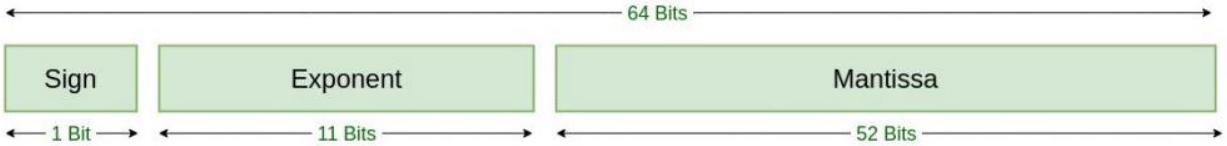
- Floating point numbers



IEEE 754 Floating Point Standards



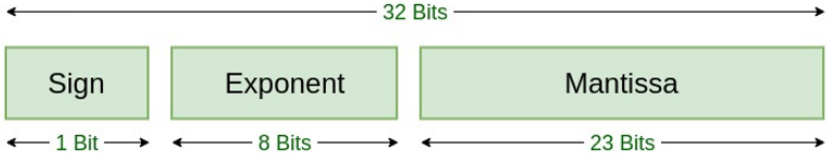
Single Precision
IEEE 754 Floating-Point Standard



Double Precision
IEEE 754 Floating-Point Standard

IEEE 754 Floating Point Standard

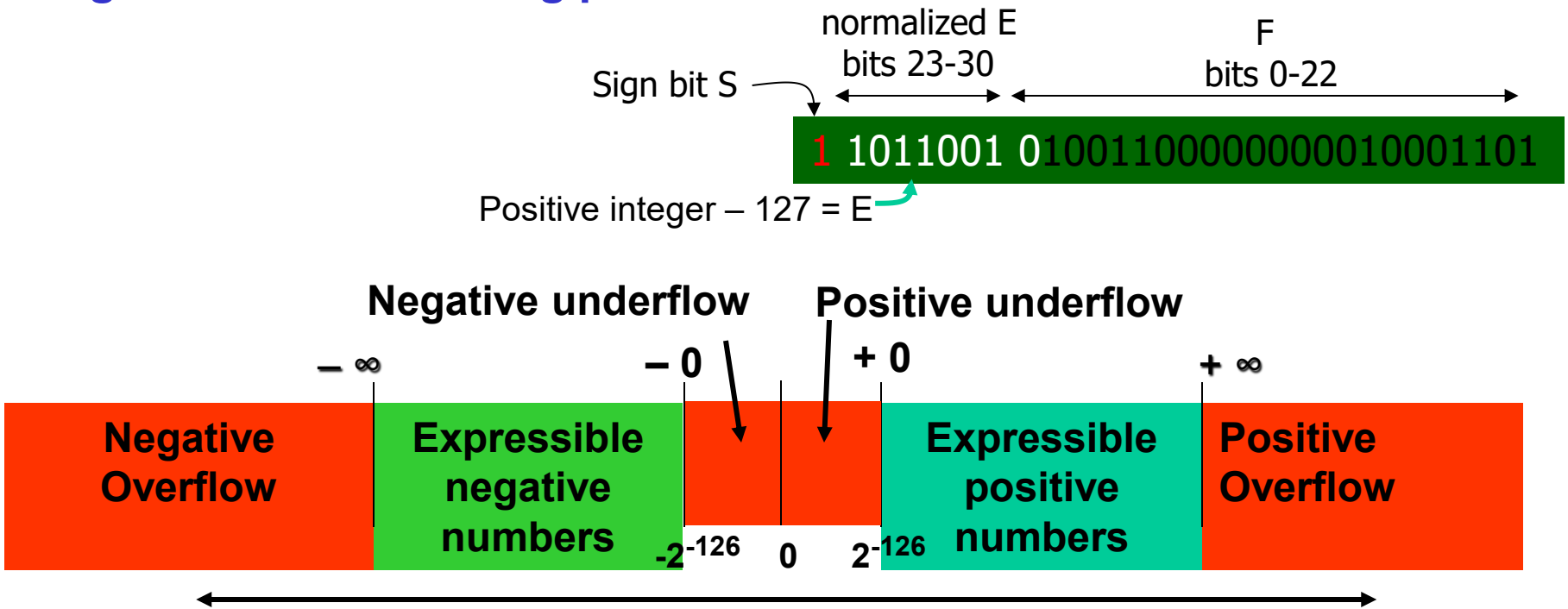
Single Precision Floating point numbers



- Biased exponent: true exponent range
  - [-126,127] is changed to [1, 254]
  - Biased exponent is an 8-bit positive binary integer.
  - True exponent obtained by subtracting 127ten or 01111111two
- First bit of significand is always 1
  - ± 1.bbbb . . . b × 2E
  - 1 before the binary point is implicitly assumed.
  - Significand field represents 23-bit fraction after the binary point.

IEEE 754 Floating Point Standard

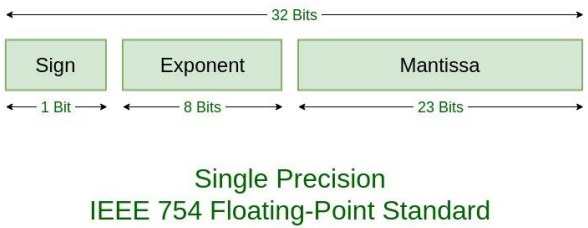
Single Precision Floating point numbers



Decimal Fraction to Binary (IEEE 754) conversion

- Represent 85.125 in IEEE 754 format
- 85 = 1010101 : 0.125 = 001
- 85.125 = 1010101.001 = 1.010101001 x 2^{6} [sign = 0]
- Biased exponent 127+6=133→= 10000101
- Normalised mantissa = 010101001 (we will add 0's to complete the 23 bits)
- The IEEE 754 Single precision is: = 0 10000101 01010100100000000000000
- Hexadecimal form 0100,0010,1010,1010,0100,0000,0000,0000

  42AA4000



Single Precision
IEEE 754 Floating-Point Standard

Binary to Decimal Fraction Conversion

Binary $(-1)^S (1.b_1b_2b_3b_4) \times 2^E$

Decimal $(-1)^S \times (1 + b_1\times2^{-1} + b_2\times2^{-2} + b_3\times2^{-3} + b_4\times2^{-4}) \times 2^E$

Example: -1.1100×2^{-2} (binary)

$= - (1 + 2^{-1} + 2^{-2}) \times 2^{-2}$

$= - (1 + 0.5 + 0.25)/4$

$= - 1.75/4$

$= - 0.4375$ (decimal)

Conversion From Hex to Decimal

- R1= 0x42220000
- **0 100 0010 0010 0010 0000 0000 0000 0000**
- **→ E'= 100 0010 0 →132 → E=132-127=5 → 2^5**
- **+ 1.0100010 x 2^5** = 101000.10 = +40.5
- R2=0xC12E0000
- **1 100 0001 0010 1110 0000 0000 0000 0000**
- **→ E'= 100 0001 0 →130 → E=130-127=3 → 2^3**
- **= - 1.0101110 x 2^3 =** 1010 = -10 + (0.5+0.25+0.125) = -10.875
- R3=0xC0800000
- **1 100 0000 1000 0000 0000 0000 0000 0000**
- **= - 1.00 x 2^2 =** 0100 = -4

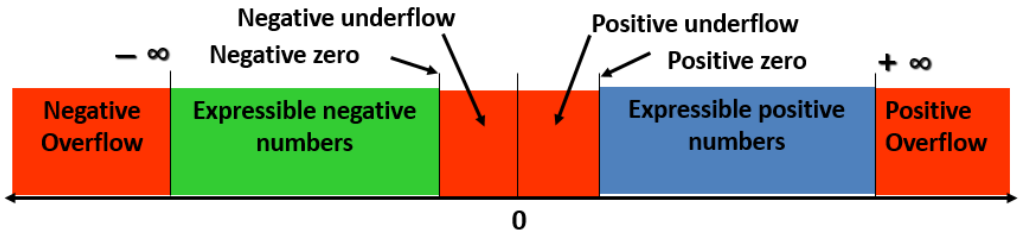
Positive Zero in IEEE 754

0 00000000 00000000000000000000000

Biased exponent

Fraction

- $+ 1.0 \times 2^{-127}$
- Smaller than the smallest positive number in single-precision IEEE 754 standard.
- Interpreted as positive zero.
- True exponent less than –126 is positive underflow

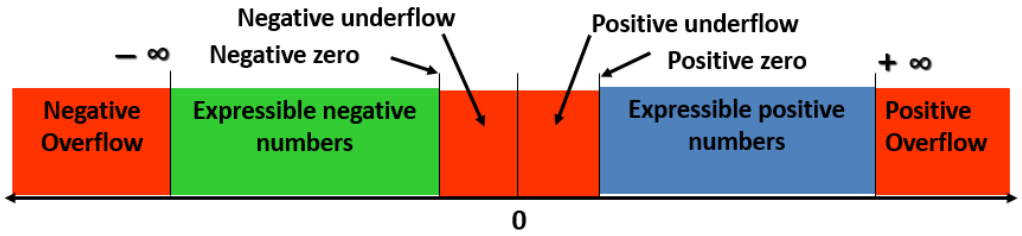


Negative Zero in IEEE 754

1 00000000 00000000000000000000000

Biased exponent | Fraction

- -1.0×2^{-127}
- Greater than the largest negative number in single-precision IEEE 754 standard.
- Interpreted as negative zero.
- True exponent less than –126 is negative underflow

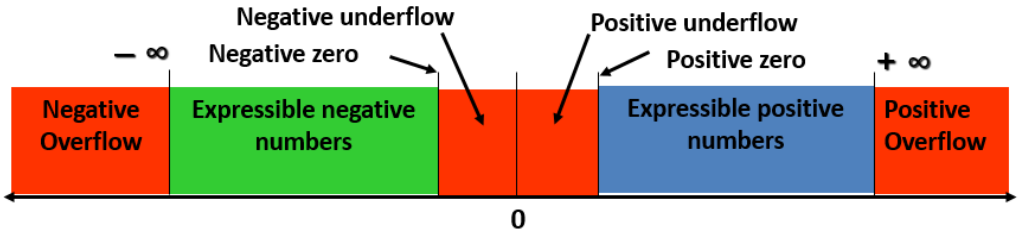


Positive Infinity in IEEE 754

0 11111111 00000000000000000000000

Biased exponent | Fraction

- $+ 1.0 \times 2^{128}$
- Greater than the largest positive number in single-precision IEEE 754 standard.
- Interpreted as $+ \infty$
- If true exponent > 127, then the number is greater than ∞. It is called “not a number” or NaN and may be interpreted as ∞.



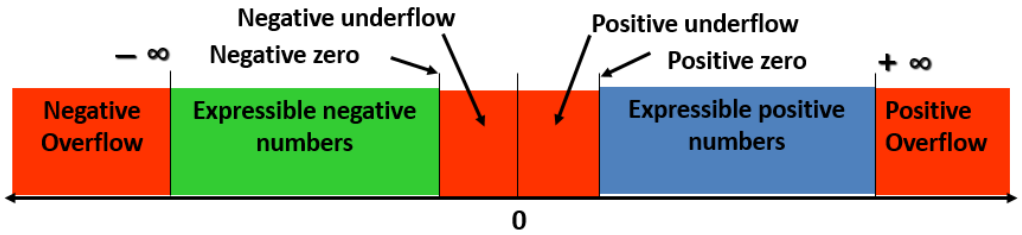
Negative Infinity in IEEE 754

1 11111111 00000000000000000000000

Biased exponent

Fraction

- -1.0×2^{128}
- Smaller than the smallest negative number in single-precision IEEE 754 standard.
- Interpreted as - ∞
- If true exponent > 127, then the number is less than - ∞. It is called “not a number” or NaN and may be interpreted as - ∞.



IEEE 754 Calculations

- Consider four floating point numbers K, L, M, N, represented as per IEEE 754 format in hexadecimal notation as follows. K= 0xC2F00000, L=0x42C80000, M= 0xC1C80000 and N=0x42200000. Which of the following statements is/are true?

  (a) K = -4*N

  (b) L = 4*M

  (c) (K+N) > L

  (d) (K+L) < (M+N)

IEEE 754 Calculations

- K= 0xC2F00000 = 11000010111100000000000000000000 = -120
- L= 0x42C80000 = 01000010110010000000000000000000 = +100
- M= 0xC1C80000 = 11000001110010000000000000000000 = -25
- N=0x42200000 = 01000010001000000000000000000000 = +40
- Which of the following statements is/are true?

  (a) K = -4*N

  (b) L = 4*M

  (c) (K+N) > L

  (d) (K+L) < (M+N)

FP Addition and Subtraction

1. **Significand alignment:** Right shift significand of smaller exponent until two exponents match.

2. **Addition:** Add significands and report error if overflow occurs.

   If significand = 0, return result as 0.

3. **Normalization**

   -Shift significand bits to normalize.

   - report overflow or underflow if exponent goes out of range.

4. **Rounding**

Example (4 Significant Fraction Bits)

- Subtraction: $0.5_{ten} - 0.4375_{ten}$
  - Floating point numbers to be added

    $1.000_{two} \times 2^{-1}$ and $-1.110_{two} \times 2^{-2}$
  - Significand of lesser exponent is shifted right until exponents match

    $-1.110_{two} \times 2^{-2} \rightarrow -0.111_{two} \times 2^{-1}$
  - Add significands, $1.000_{two} + (-0.111_{two})$

    Result is $0.001_{two} \times 2^{-1}$
  - Normalize, $1.000_{two} \times 2^{-4}$

    No overflow/underflow since

    $127 \geq \text{exponent} \geq -126$

    $1.000_{two} \times 2^{-4} = (1+0)/16 = 0.0625_{ten}$

```
 01000
+11001
 00001
```

2's complement addition, one bit added for sign

FP Multiplication

1. Separate sign
2. Add exponents (integer addition)
3. Multiply significands (integer multiplication)
4. Normalize, round, check overflow/underflow
5. Replace sign



johnjose@iitg.ac.in

http://www.iitg.ac.in/johnjose/

https://www.linkedin.com/in/john-jose-bb8431106/